

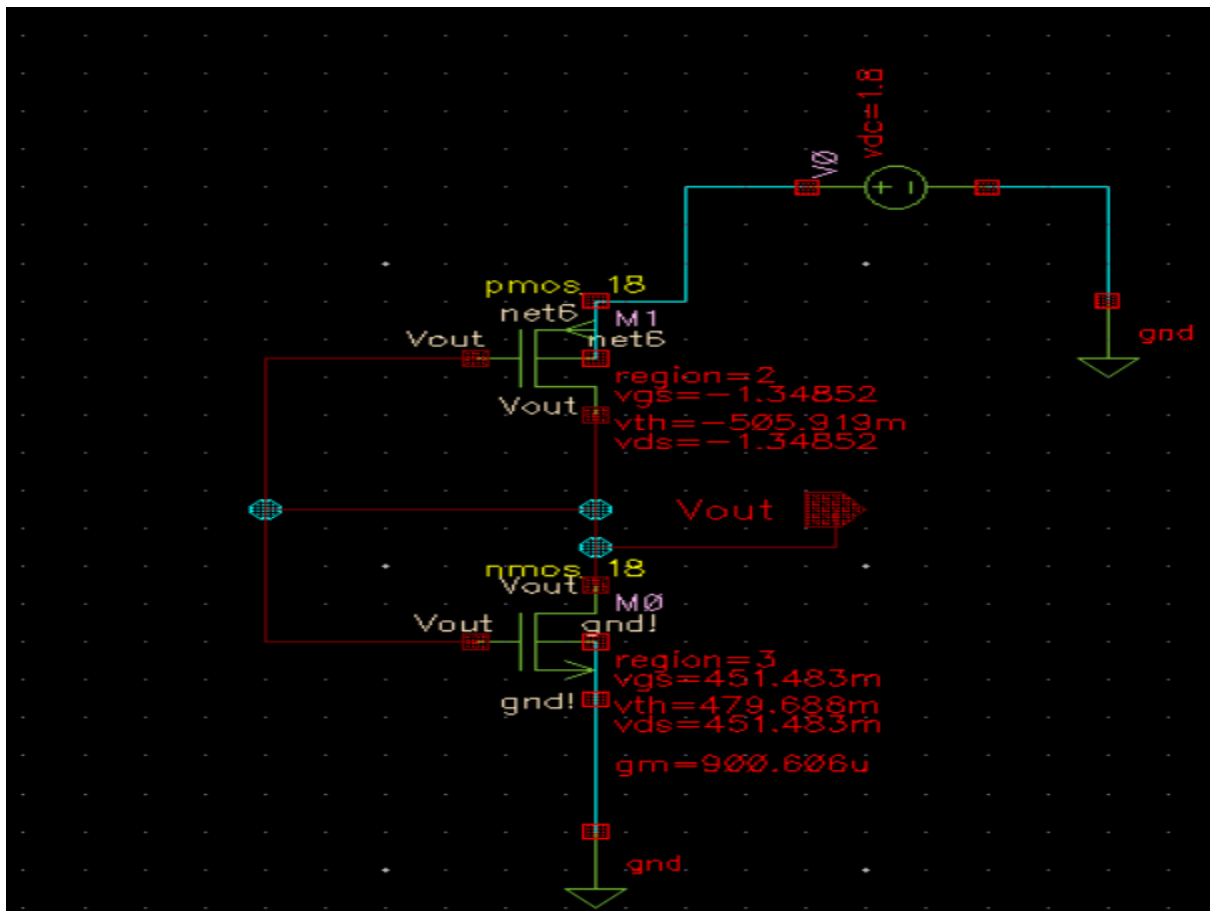
CMOS ANALOG IC DESIGN

ASSIGNMENT_2

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TASK :-1

SCHEMATIC OF VOLTAGE DIVIDER (0.25VDD)



In this circuit nmos is in sub threshold condition because vgs is 0.451V and the Vtn is 0.479V and transistor in triode act like a resistor and leakage current flows through the circuit .And above PMOS will operate in saturation.

To find the W/L ratio we need to equate the current of both the transistors and we will get the W/L ratio and making length as constant and varying the width of both the transistors and able to get Vout as 0.45V.

$\mu_n C_{ox} = 65 \text{ mA/V}^2$

$$\mu_{p_Cox} = 250 \text{ mA/V}^2$$

Ln(uM)	Wn(uM)	Lp(uM)	Wp(uM)
0.18	30	0.18	0.42

$$\frac{1}{2} \mu_{p_Cox} \left(\frac{W}{L} \right)_p (V_{fg} - V_{tp})^2 = \frac{1}{2} \mu_{n_Cox} \left(\frac{W}{L} \right)_n (V_{gs} - V_{tn})^2$$

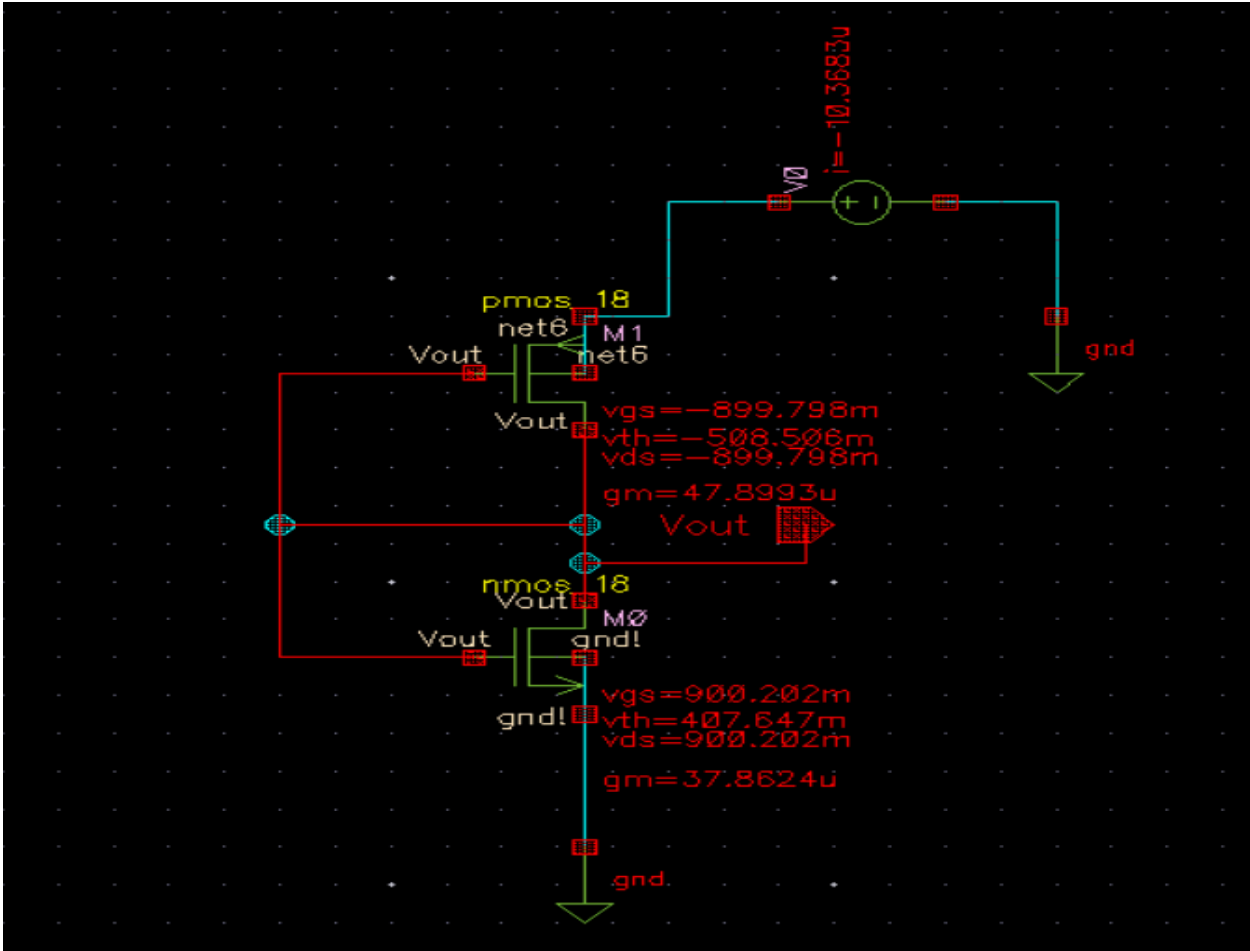
$$65 \left(\frac{W}{L} \right)_p (1.8 - 0.45 - 0.51)^2 = 250 \left(\frac{W}{L} \right)_n (0.45 - 0.48)^2$$

$$\frac{\left(\frac{W}{L} \right)_p}{\left(\frac{W}{L} \right)_n} = \frac{250 (0.45 - 0.48)^2}{65 (1.8 - 0.45 - 0.51)^2}$$

$$\frac{\left(\frac{W}{L} \right)_n}{\left(\frac{W}{L} \right)_p} = 204.08$$

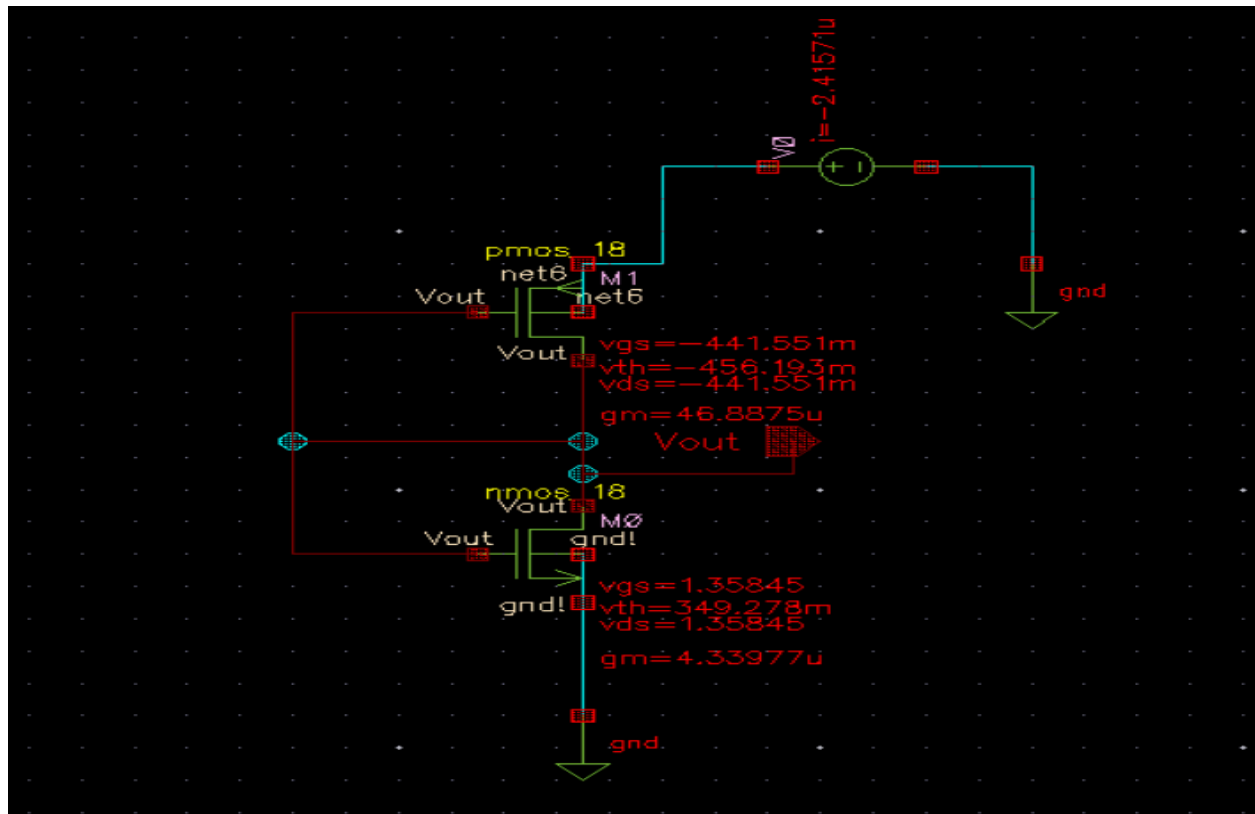
$$\boxed{\left(\frac{W}{L} \right)_n = 204.08 \cdot \left(\frac{W}{L} \right)_p}$$

SCHEMATIC OF VOLTAGE DIVIDER (0.5VDD)



Ln(uM)	Wn(uM)	Lp(uM)	Wp(uM)
0.66	0.22	0.18	0.42

SCHEMATIC OF VOLTAGE DIVIDER (0.75VDD)



Ln(uM)	Wn(uM)	Lp(uM)	Wp(uM)
10	0.22	0.22	6

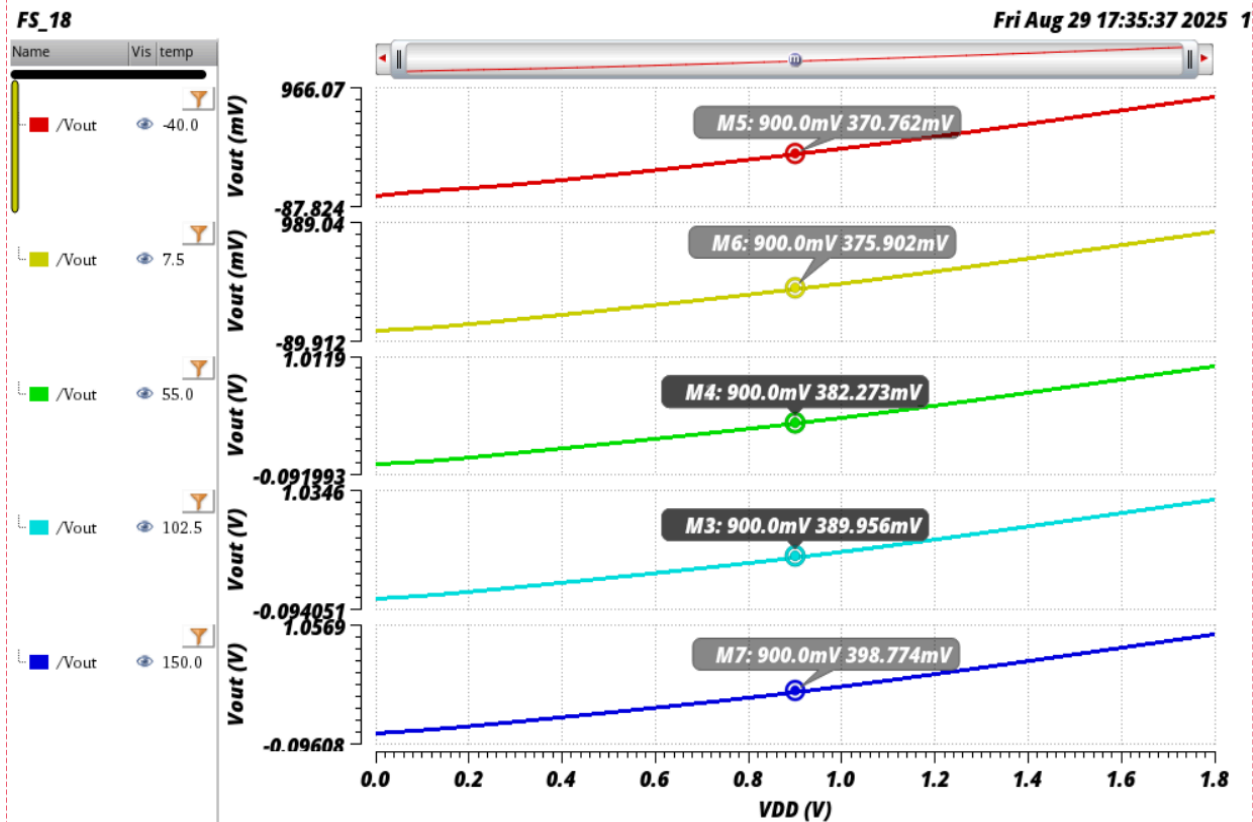
FF: Fast-Fast (both transistors are fast).

SS: Slow-Slow (both transistors are slow).

TT: Typical-Typical (both transistors are average).

FS: Fast-Slow (first transistor is fast, second is slow).

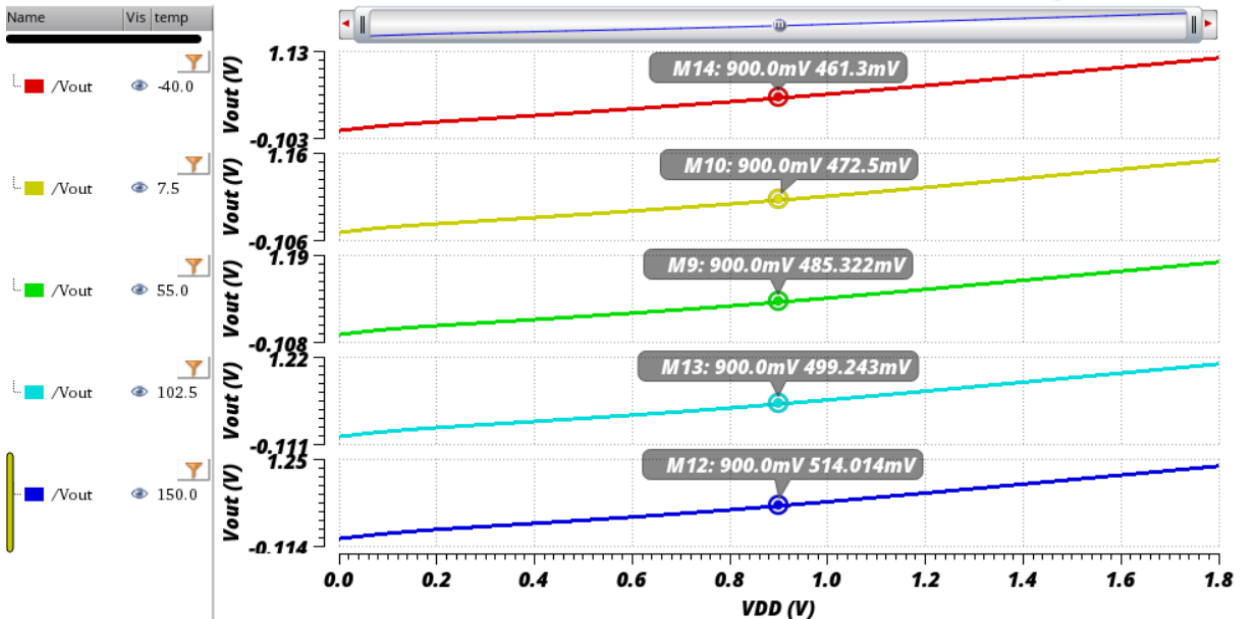
SF: Slow-Fast (first transistor is slow, second is fast).



In FS_18 as temperature increases the V_{out} is also increasing at constant operating because of the transistor properties, like threshold voltage, shift with temperature. mobility is dependent on temperature. The increase is slightly low compared to FF_18. Percentage change (-40°C to 150°C): 7.56%

FF_18

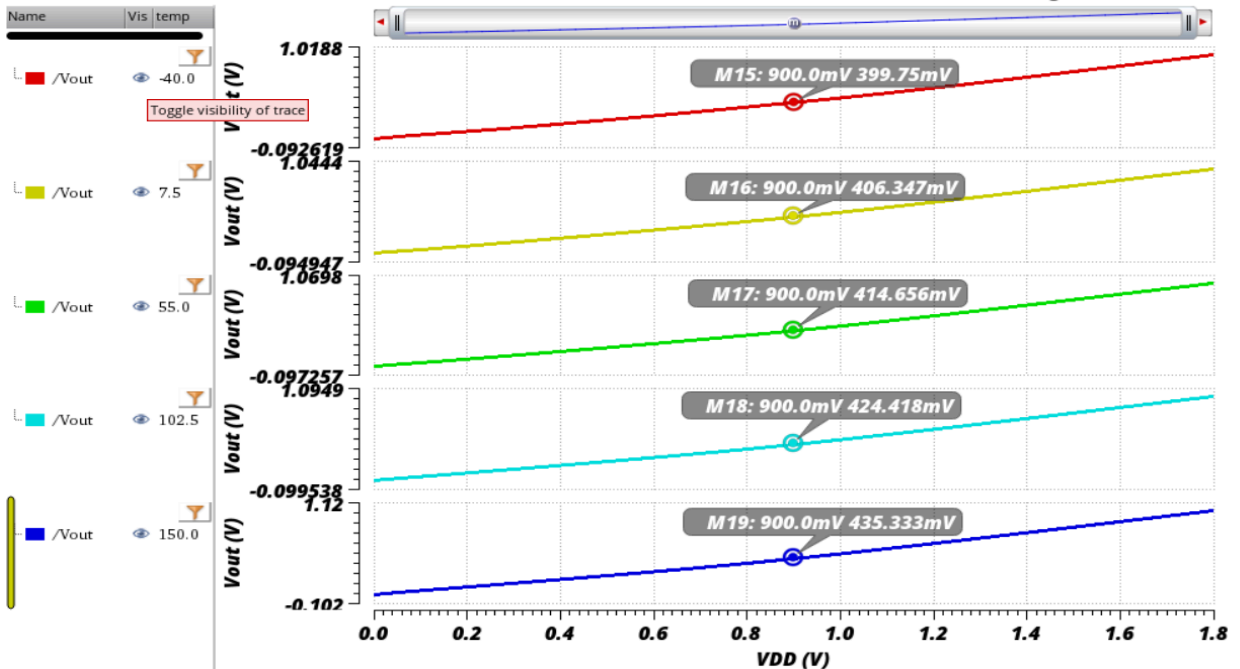
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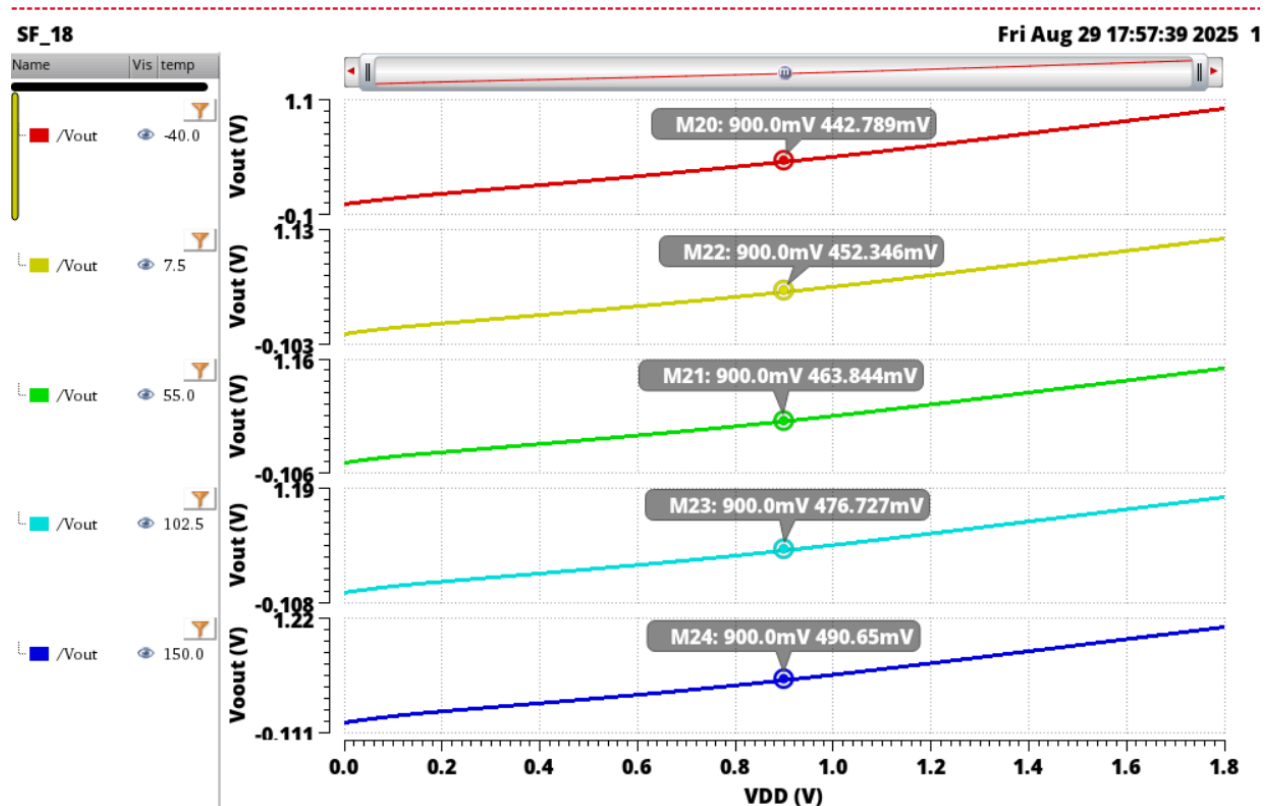
In FF_18 as temperature increases the V_{out} also increases and current flows more make the output high at least time. because mobility increases and increases linearly and high compared to FS_18. Percentage change (-40°C to 150°C): 11.43%.

TT_18

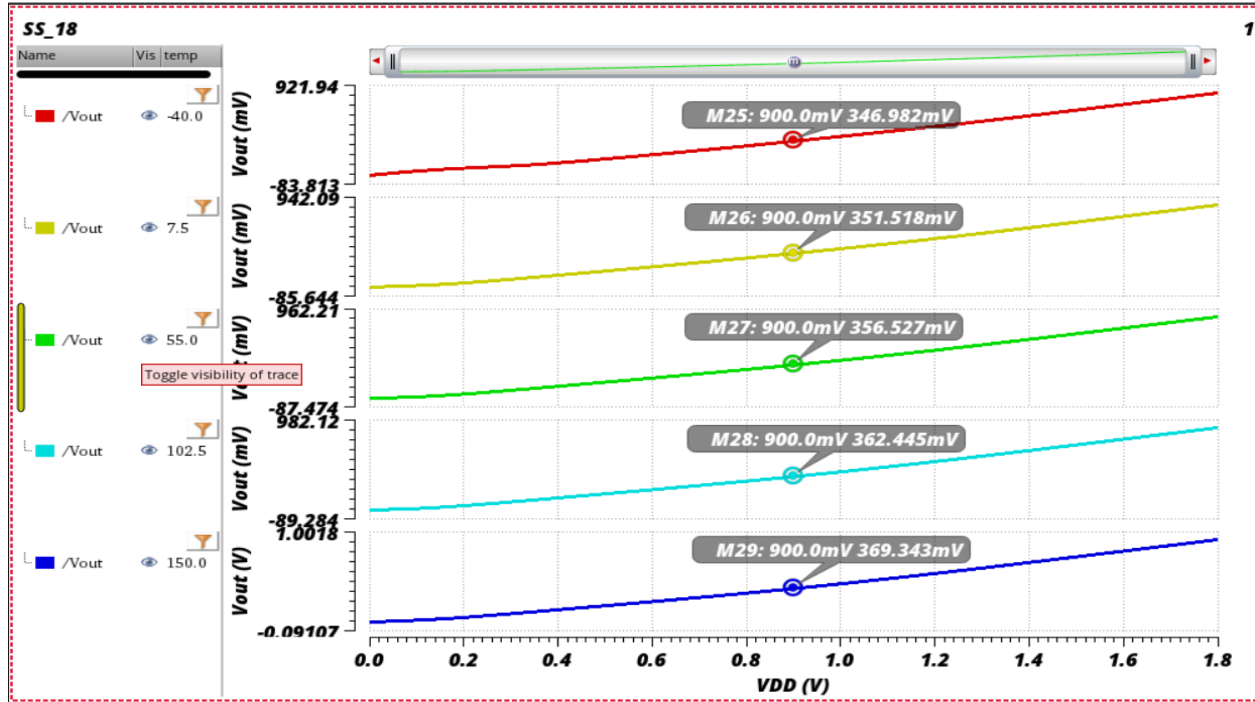
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In this also the Vout increases linearly as temperature increases but the magnitude is small compared to the FF_18 ,SF_18 it operates moderately.Percentage change (-40°C to 150°C): 8.90%



In this also the increases is linear and increases with temperature and very small.Percentage change (-40°C to 150°C): 10.81%



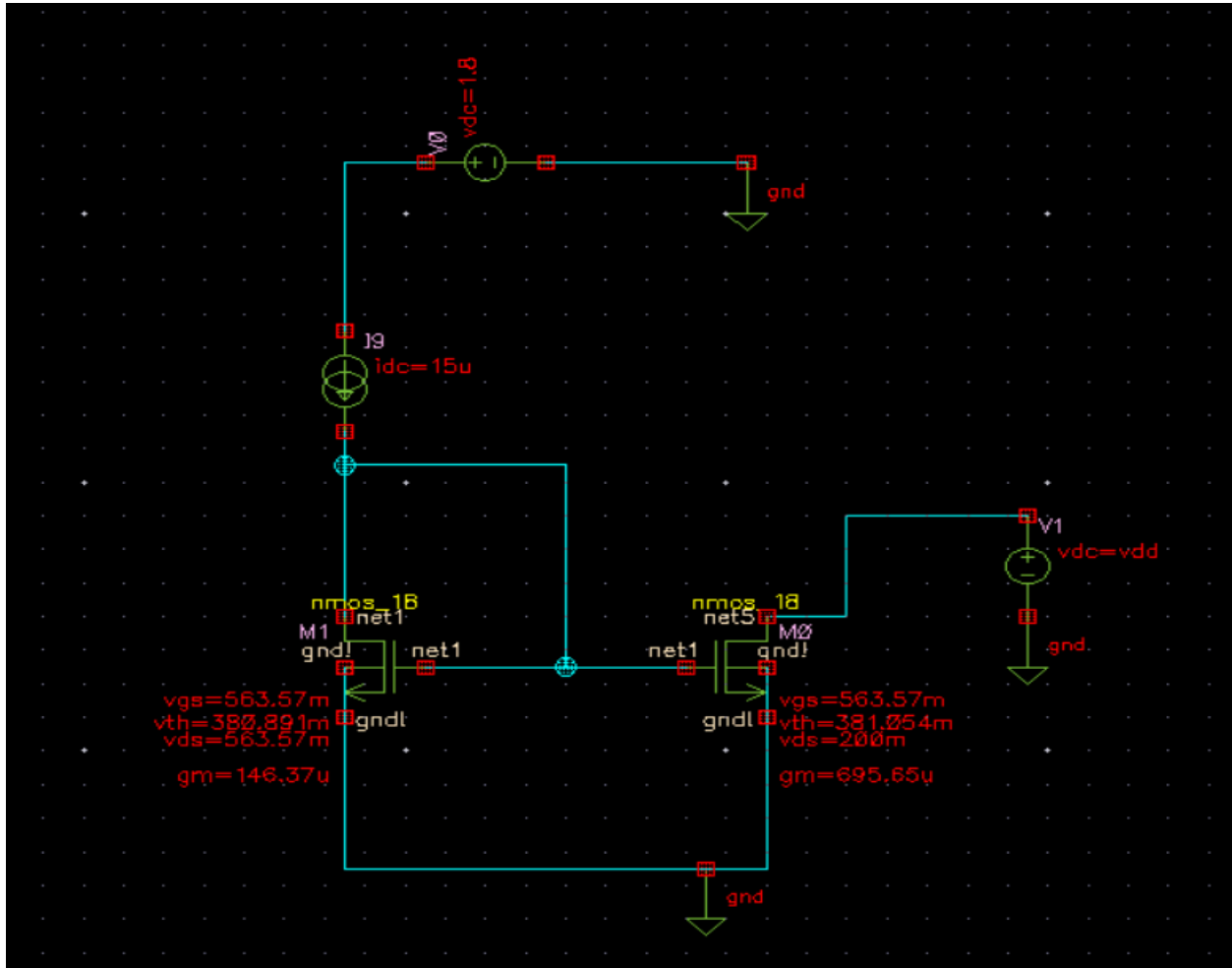
In this as the temperature increases the increase is very small compared to all other process corners. Percentage change (-40°C to 150°C): 6.44%

The table below shows the process corners for various temperature

TEMP C	FS_18 (mV)	FF_18 (mV)	TT_18 (mV)	SF_18 (mV)	SS_18 (mV)
-40	370.762	461.3	399.75	442.789	346.982
7.5	375.902	472.5	406.347	452.346	351.518
55	382.273	485.322	414.656	463.844	356.527
102.5	389.956	499.243	424.418	476.727	362.445
150	398.774	514.014	435.33	490.65	369.343

TASK:-2

SCHEMATIC OF SIMPLE CURRENT MIRROR



Current Mirror Basics: In a cascode current mirror, I_{out} should equal $75\ \mu A$ in this case as long as both transistors (M1, M2) remain in saturation. The cascode structure improves output impedance and current matching.

Saturation Condition: For an NMOS transistor to be in saturation, $V_{ds} \geq V_{gs} - V_{th}$, where V_{th} is the threshold voltage. For M2, V_{ds2} is the voltage at its drain (swept from 0 to VDD), and V_{gs2} is set by the reference current I_{ref} through M2.

$I_{ref} = 15\ \mu A$ (current source).

$I_{out} = 75 \mu A$ (desired output current.)

$$I_D = (1/2) * \mu * C_{ox} * (W/L) * (V_{gs} - V_{tn})^2$$

$$I_{out}/I_{ref} = (W/L)_{out}/(W/L)_{ref}$$

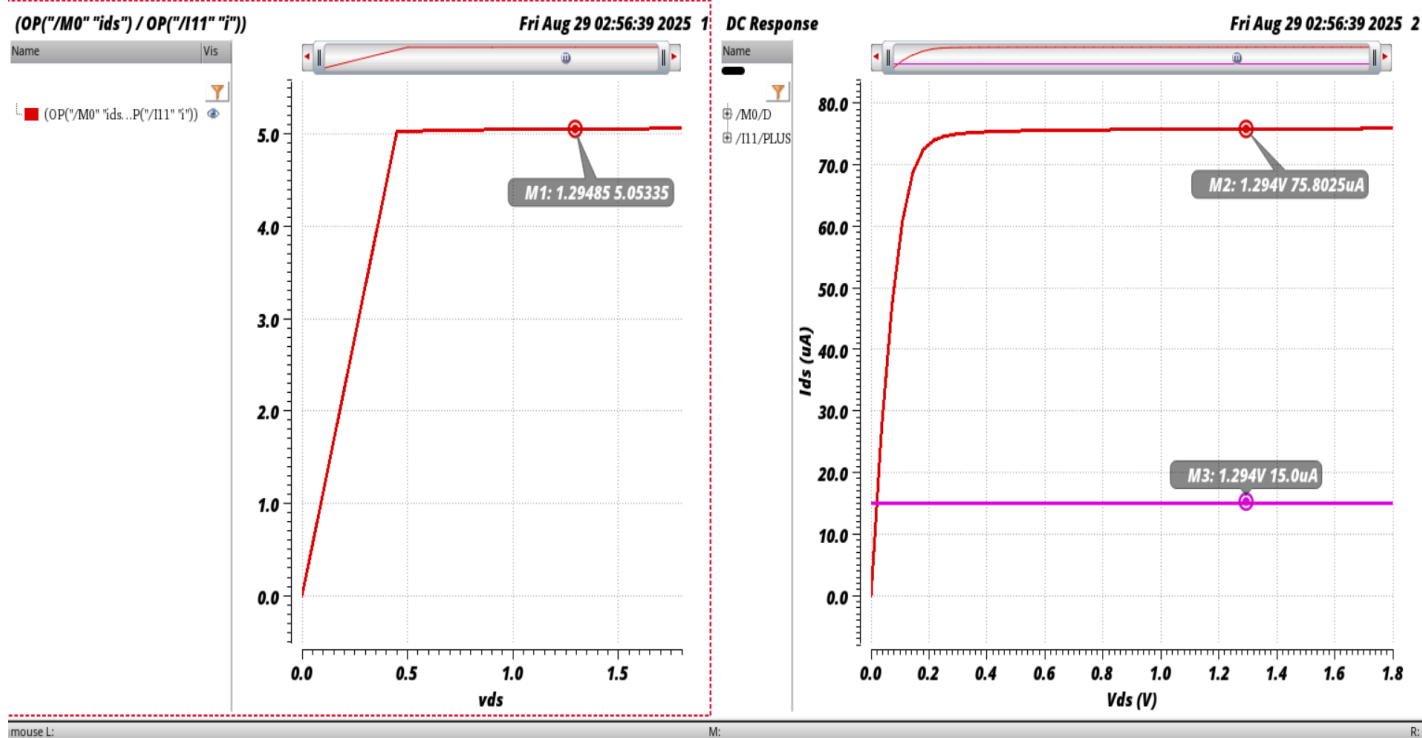
Keeping length the same and varying the width.

$$75/15 = W_{out}/W_{ref}$$

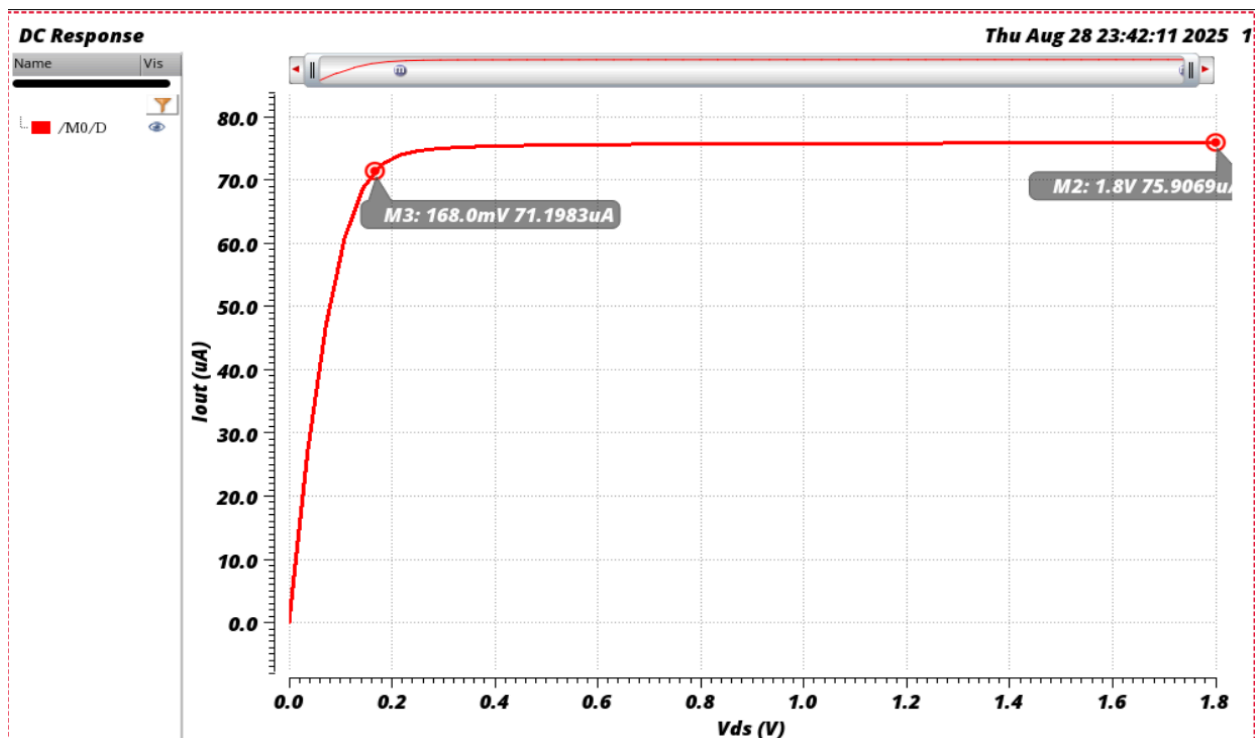
$$W_{out} = 5 W_{ref}$$

$$W_{out} = 5 * 0.42 = 2.1 \mu m$$

The width of the mosfets of M1 are kept the same .Varied the width of the second transistor.



RATIO OF I_{out}/I_{ref} OF CURRENT MIRROR



ERROR AT EDGE OF SATURATION AND SATURATION

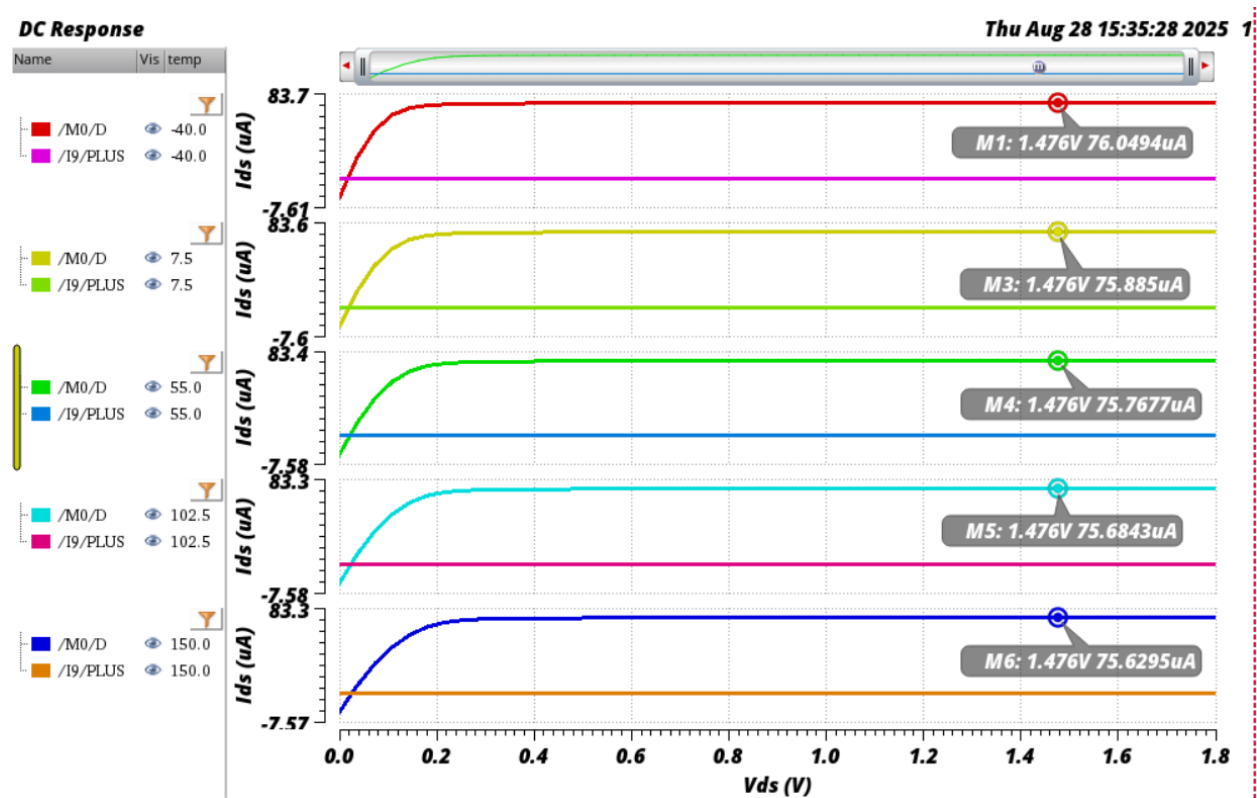
The minimum voltage to keep the transistors M2 in saturation is 168mV. I have done this by varying the vdd and found the region of operation.

The transistors at the edge of saturation is at 168mV and drain current is 71.198uA and at saturation of vdd at 1.8V is 75.99uA

$$\text{Error} = (75.99 - 71.81) / 75.99$$

$$= 6.3\%$$

The error is high I compared to the cascode current mirror and the output impedance of the circuit is low. The saturation point also is at low voltage.

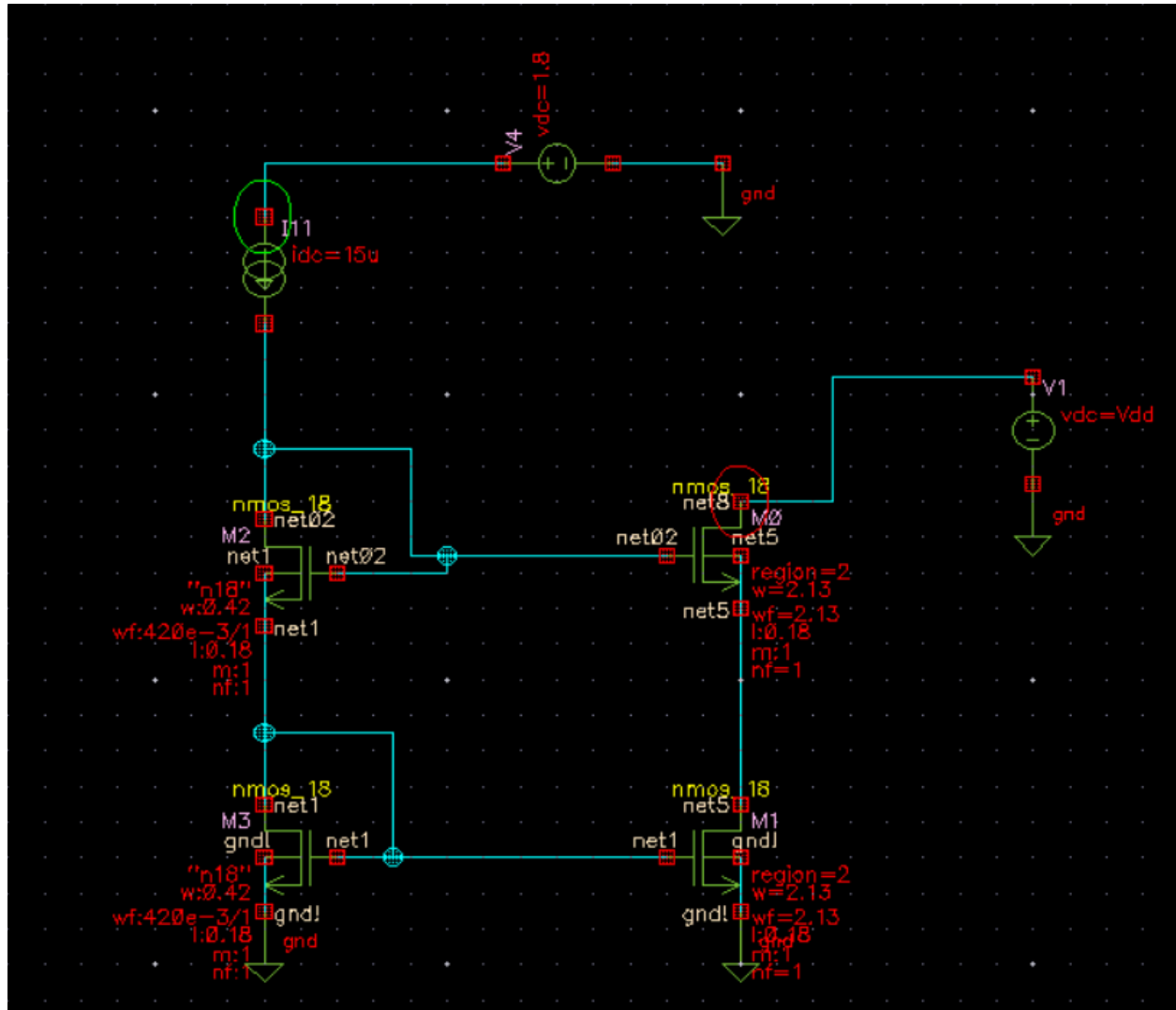


DEPENDANCE OF CURRENT ON TEMPERATURE

As the temperature increases from -40°C to 150°C , the current ratio ($I_{\text{out}} / I_{\text{ref}}$) slightly decreases because the threshold voltage (V_{tn}) of the transistors drops with higher temperatures. At lower temperatures (-40°C), the current ratio stays closer to 5, since I_{out} should be 5 times I_{ref} , showing the circuit works well with stable currents. The plot of $I_{\text{out}} / I_{\text{ref}}$ versus temperature shows a gradual decline, indicating the current mirror becomes less accurate as temperature rises, but the change is small over the range. we might need to adjust the W/L ratios to compensation to keep I_{out} close to $75\text{ }\mu\text{A}$.

TASK:-3

SCHEMATIC OF CASCODE CURRENT MIRROR



Current Mirror Basics: In a cascode current mirror, I_{out} should equal $75\mu A$ in this case as long as both transistors ($M1$, $M2$, $M3$, $M4$) remain in saturation. The cascode structure improves output impedance and current matching.

Saturation Condition: For an NMOS transistor to be in saturation, $V_{ds} \geq V_{gs} - V_{tn}$, where V_{tn} is the threshold voltage. For M3, V_{ds3} is the voltage at its drain (swept from 0 to VDD), and V_{gs3} is set by the reference current I_{ref} through M1 and M2.

$I_{ref} = 15 \mu A$ (current source).

$I_{out} = 75 \mu A$ (desired output current.)

All transistors are matched (same process parameters: mobility μ , oxide capacitance C_{ox} , and threshold voltage V_T).

L is fixed ($L = 0.18 \mu m$)

The current is proportional to $(W/L) \times (\mu C_{ox}/2) \times (V_{GS} - V_T)^2$ in saturation.

The cascode structure requires M3 and M2 to have sufficient V_{ds} to stay in saturation.

The drain current in saturation for an NMOS transistor is given by:

$$I_D = (1/2) * \mu * C_{ox} * (W/L) * (V_{gs} - V_{tn})^2$$

For the reference branch (M1 and M2):

$I_{ref} = 15 \mu A$ is set by the current source through M1 (diode-connected) and mirrored by M2.

Let the W/L ratio of M1 and M2 be $(W_1/L_1) = (W_2/L_2)$.

For the output branch (M3 and M4):

$I_{out} = 75 \mu A$ is the desired current through M3 and M4.

Let the W/L ratio of M3 and M4 be $(W_3/L_3) = (W_4/L_4)$

Since $I_{out}/I_{ref} = 75 \mu A / 15 \mu A = 5$, the W/L ratio of the output branch should be 5 times that of the reference branch (made over drive voltage to be the same).

$$I_{out}/I_{ref} = (W/L)_{out} / (W/L)_{ref}$$

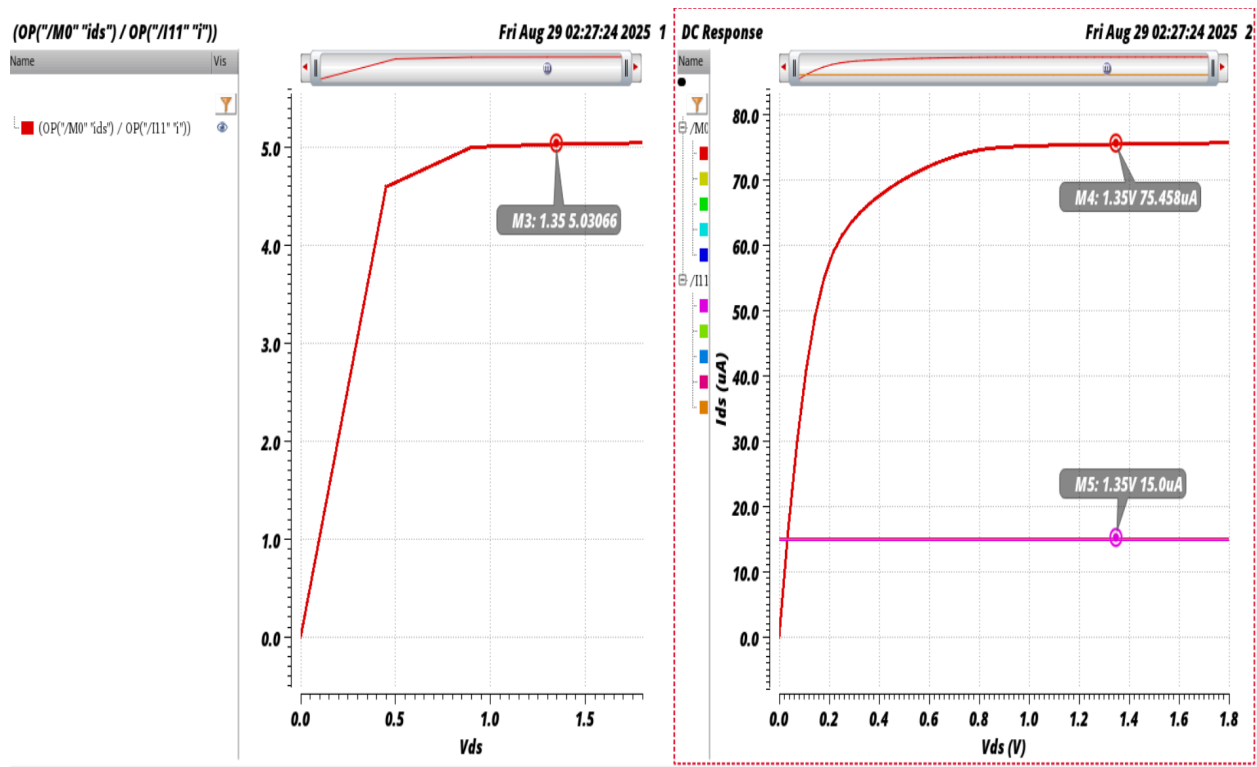
Keeping length the same and varying the width.

$$75/15 = W_{out}/W_{ref}$$

$$W_{out} = 5 W_{ref}$$

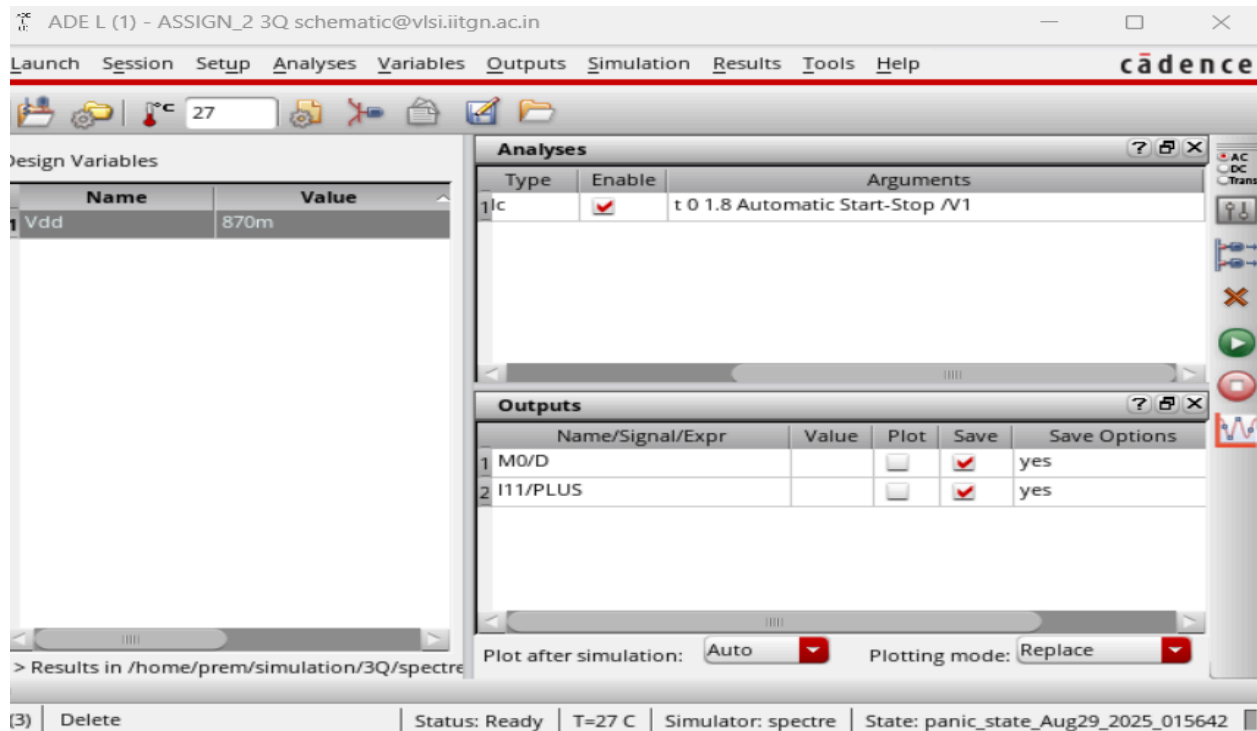
$$W_{out} = 5 \times 0.42 = 2.1 \mu\text{m}$$

The width of the mosfets of M3 and M4 are kept the same .

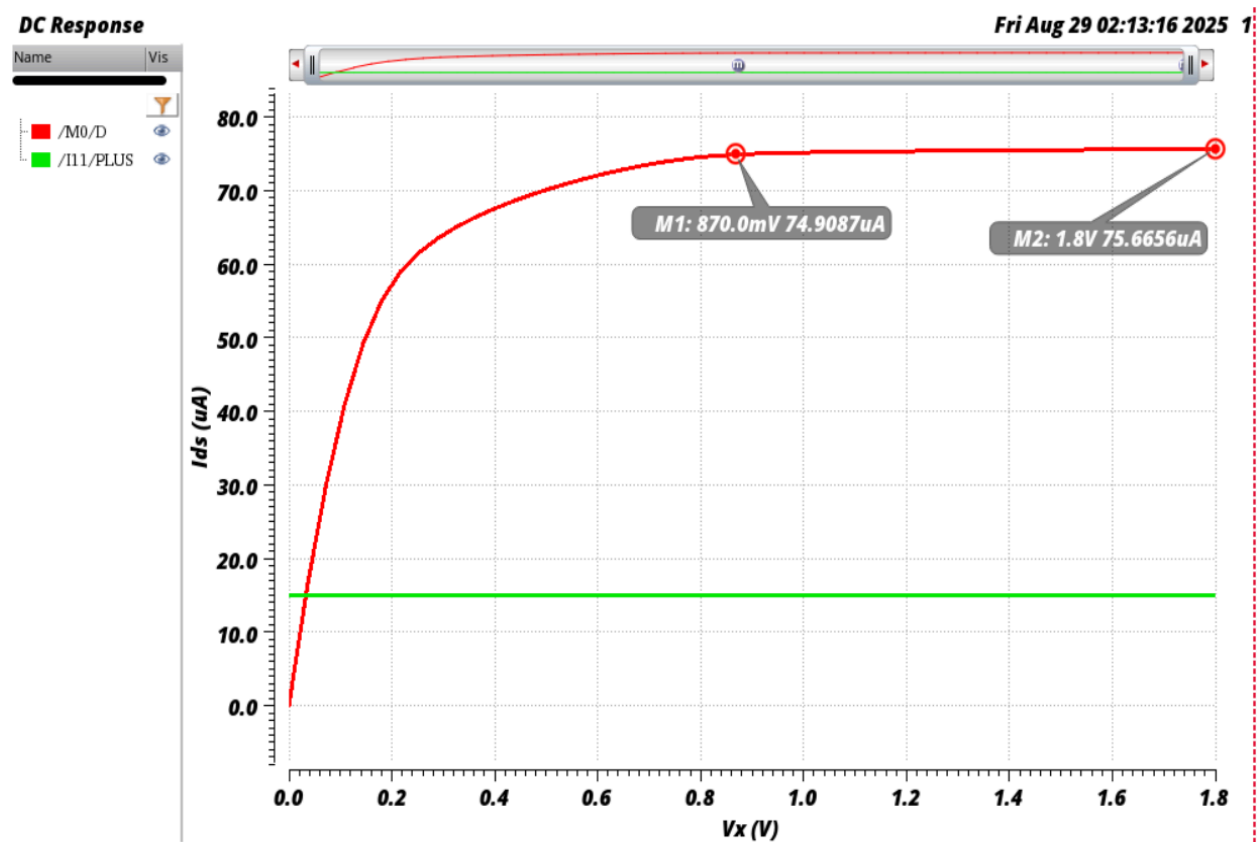


RATIO OF I_{out}/I_{ref} CASCODE CURRENT MIRROR

The minimum voltage to keep the transistors M3 and M4 in saturation is 870mV i have done this by varying the vdd and found the region of operation



EDGE OF OPERATION FROM LINEAR TO SATURATION



ERROR AT SATURATION AND LINEAR REGION

The transistors at the edge of saturation is at 870mV and drain current is 74.9uA and at saturation of vdd at 1.8V is 75.66uA

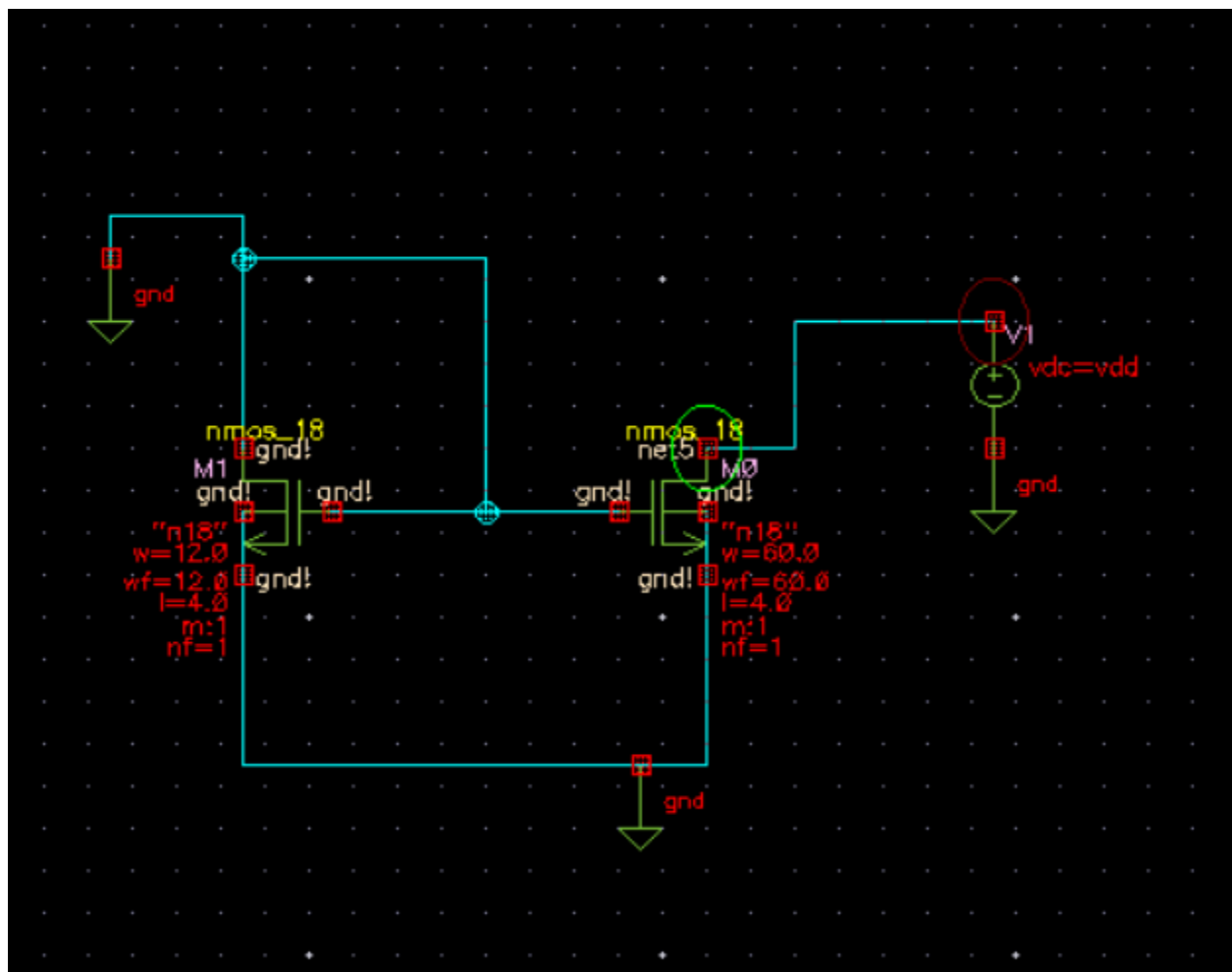
$$\text{Error} = (75.66 - 74.9) / 75.66$$

$$= 1.004\%$$

The error is very small compared to the simple current mirror and the output impedance of the circuit is high.

TASK:-4

SCHEMATIC OF SIMPLE CURRENT MIRROR Rout

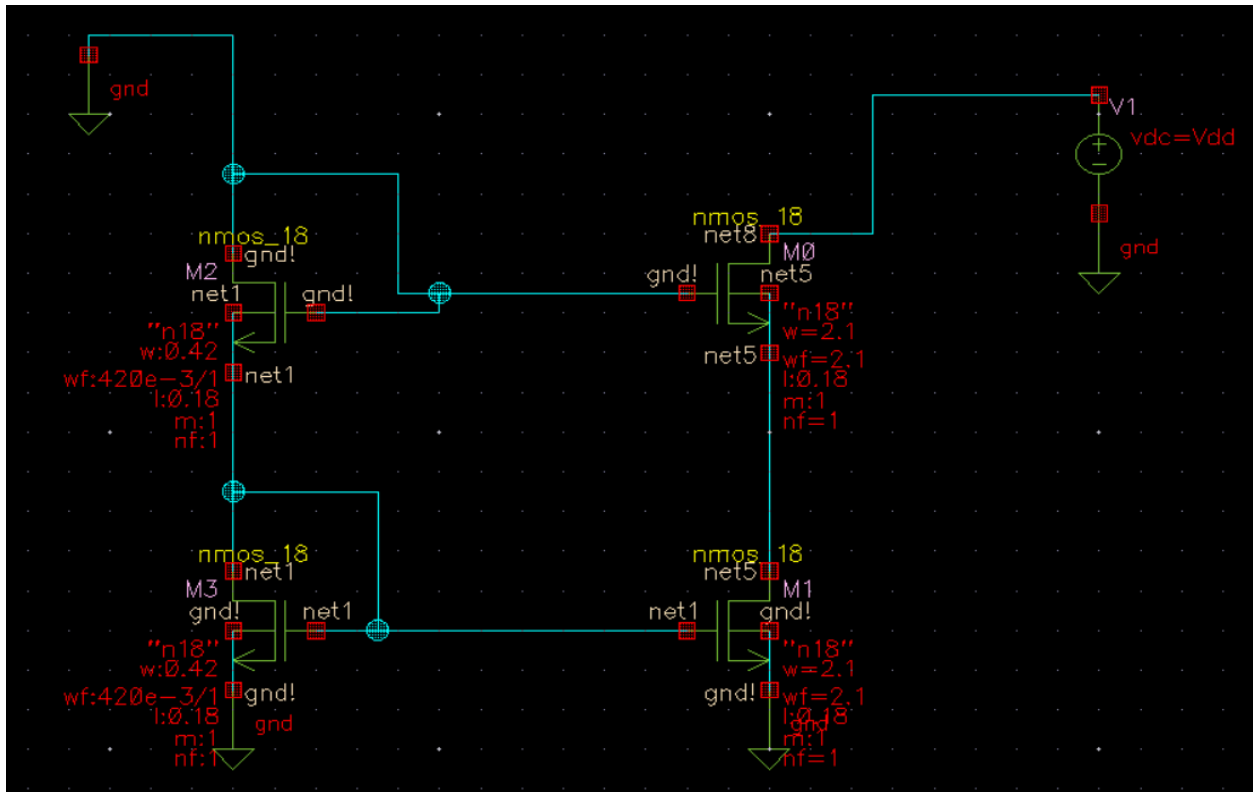


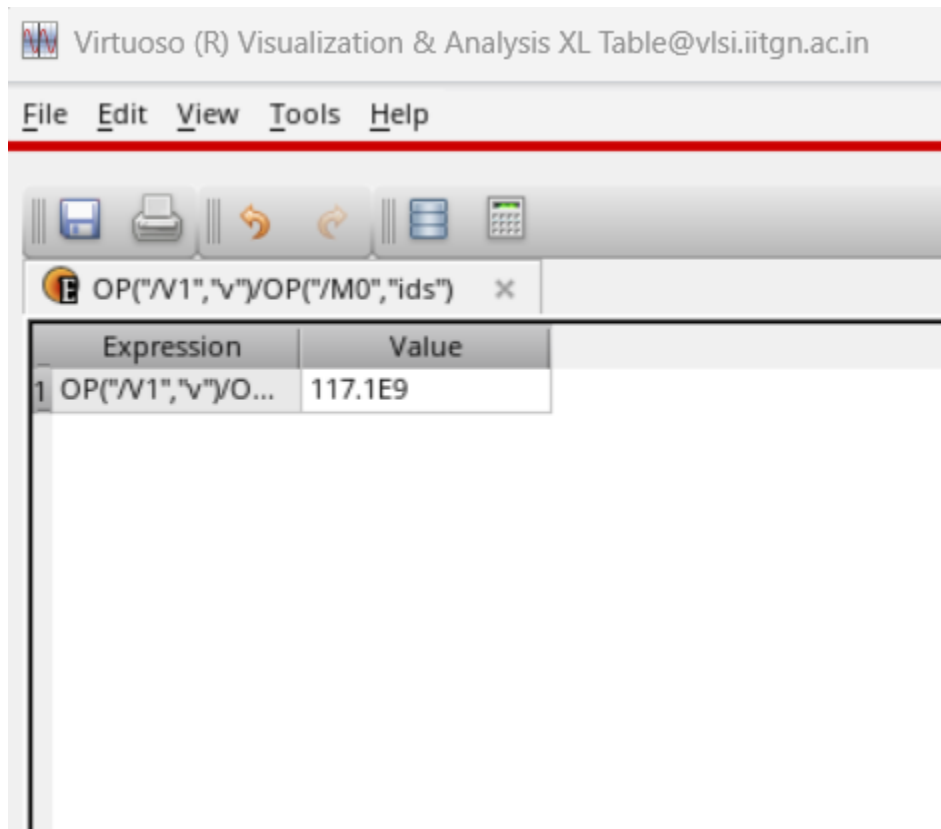
Virtuoso (R) Visualization & Analysis XL Table@vlsi

Expression	Value
1 OP("/V1","v")/OP("/M0","ids")	3.943E9

IMPEDANCE OF SIMPLE CURRENT MIRROR CIRCUIT

SCHEMATIC OF CASCODE CURRENT MIRROR Rout





OUTPUT IMPEDANCE OF CASCODE CIRCUIT

SIMPLE CURRENT MIRROR	CASCODE CURRENT MIRROR
Simple design	Complex design and symmetric
Threshold voltage is low for going into saturation	Threshold voltage is igh for going into saturation
Error percentage is high	Error percentage is low
Output impedance is low	Output impedance is high
Channel length modulation reduces precision	Very low dependency on channel length and precise
Power Consumption is low	Power Consumption is high